2. Requirement Analysis Phase

2.1 Introduction

The Requirement Analysis Phase identifies, analyzes, and documents the requirements of PocketSmart AI: Your Smart Budget & Recommendation Assistant. It focuses on understanding the problems faced by users in managing money and defining the features needed to solve them. PocketSmart AI is designed as an intelligent financial assistance system that uses Google Gemini to analyze spending, plan budgets, answer money questions, and recommend smarter choices.

2.2 Problem Analysis

Personal budgeting can be difficult. Users may not know where their money goes, may overspend without noticing, may not know how much to save, and may find it hard to compare options before buying. Traditional tools are generally static and do not adapt to the individual. The major problems identified are:

- Lack of awareness about spending habits.
- Difficulty in creating a realistic, personalized budget.
- Time-consuming manual categorization and analysis of expenses.
- Poor savings planning for short-term and long-term goals.
- Difficulty in choosing cost-effective products or alternatives.
- Lack of simple, immediate answers to everyday money questions.
- Difficulty in tracking financial progress over time.

2.3 Objectives of Requirement Analysis

- To identify the needs and expectations of users.
- To define the major functionalities of PocketSmart AI.
- To identify the required hardware and software resources.
- To determine functional and non-functional requirements.
- To define requirements for Gemini-based AI integration.
- To provide a clear foundation for design and development.

2.4 Functional Requirements

2.4.1 User Registration and Authentication

- User registration and secure login.
- User profile management (name, currency, monthly income).
- Logout functionality.
- Basic authentication and input validation.

2.4.2 Expense and Income Tracking

- Add, edit, and delete income and expense entries.
- Record amount, category, date, and notes.
- Optional AI-suggested category from the expense description.
- Filter entries by date and category.

2.4.3 AI Budget Planner

- Accept monthly income, fixed expenses, and savings goals.
- Generate a category-wise monthly budget using Gemini.
- Compare actual spending against the budget.

2.4.4 Spending Insights

- Analyze recorded expenses and generate a plain-language summary.
- Highlight overspending categories and saving opportunities.

2.4.5 Savings Goals

- Accept goal name, target amount, and target date.
- Generate a suggested monthly or weekly saving plan.
- Track progress towards each goal.

2.4.6 Smart Recommendations

- Accept a purchase need and budget limit.
- Suggest budget-friendly options, alternatives, and cost-saving tips.
- Explain the reasoning behind each recommendation.

2.4.7 AI Finance Chat

- Accept money-related questions in natural language.
- Maintain conversation context for follow-up questions.

2.4.8 History and Dashboard

- Store previous budgets, insights, recommendations, and chats.
- Display income vs expenses, category breakdown, and savings trends through charts.

2.5 Non-Functional Requirements

Requirement	Description
Usability	Simple, clean, and responsive interface usable without financial or technical knowledge.
Performance	AI responses and page loads should appear within a reasonable time.

Requirement	Description
Security	Secure authentication, protected APIs, safe storage of financial data, and the Gemini API key kept on the server.
Privacy	Users can access only their own financial records; no bank credentials are collected.
Reliability	Accurate storage of data and graceful handling of API errors or timeouts.
Scalability	Architecture should allow new categories, features, and users to be added.
Maintainability	Modular, well-organized code that is easy to test and extend.

2.6 Hardware Requirements

- Computer or laptop with minimum 4 GB RAM and 10 GB free storage.
- Internet connection (required for Gemini API access).
- Keyboard and mouse; smartphone or computer for accessing the application.

2.7 Software Requirements

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python, FastAPI
Database	SQLite / PostgreSQL
AI / LLM	Google Gemini API
Authentication	JWT
Charts	Chart.js
IDE	Visual Studio Code
Version Control	Git and GitHub
Operating System	Windows / Linux / macOS

2.8 Expected Outcome

The phase produces a well-defined set of requirements that serves as the foundation for the planning, design, development, testing, and demonstration phases. PocketSmart AI is expected to give users a convenient platform for understanding their spending, planning budgets, reaching savings goals, and making smarter purchase decisions with AI support.